

“CS - 125” SYSTEMS GUIDE



An overview of the instrumentation and communication systems onboard CS-125

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OVERVIEW OF ONBOARD SYSTEMS

The following table gives a brief overview of the primary electrical systems onboard CS-125

System	Description
<i>Instrumentation</i>	<i>FARO System</i>
<i>Navigation</i>	<i>PC based (Adrena, Expedition & Deckman)</i>
<i>Comms</i>	<i>B&G V90S VHF</i>
<i>Load Monitoring</i>	<i>Future Fibres & Harken Load Pins</i>
<i>Inertial Navigation System</i>	<i>IXblue Quadrans</i>
<i>Satellite Comms</i>	<i>KVH V7 HTS & Iridium Thales VesseLink</i>
<i>IT Network</i>	<i>Peplink</i>

POWER DISTRIBUTION

FUSES & BREAKERS

Osprey supplied DC systems are powered by three DC distribution boxes. These are located on electronic equipment panels in the fwd lazarette bulkhead, nav station and the mast base (stbd side outside of the engine room).

Details of each breaker located in the different distribution boxes can be seen below:

FWD LAZ BULKHEAD DC DISTRIBUTION BOX

Breaker ID / Rating	System
F81 / 2A	AIS Transponder
F82 / 10A	Iridium Thales VesseLINK
F83 / 4A	Pepwave HD2 Router
F84 / 2A	Pilot CAN Bus Supply
F85 / 4A	B&G Halo Radar
F86 / 2A	Instruments (AFT CAN Bus Supply)

NAVIGATION STATION DC DISTRIBUTION BOX

Breaker ID / Rating	System
F71 / 10A	Instruments (FARO Aft, Novatel GPS, 3DM, CAN Supply)
F72 / 10A	Displays (ATP1 CPU, A+T Ethernet Boxes & Loadcell Amplifiers)
F73 / 10A	Chargers (BC214 VHF Charger & Toughbook)
F74 / 6A	Laptop
F75 / 16A	LAN (SD Switch#1, Switch #2 & MOXA NPort)

F76 / 2A	AIS ALARM
F77 / 4A	PC's
F78 / 4A	Monitors
F79 / 4A	Intercom
F81 / 6A	Jon Buoy

MAST BASE DC DISTRIBUTION BOX

Breaker ID / Rating	System
F91 / 2A	Instruments (FARO Fwd)
F92 / 4A	A+T Ethernet (Fwd A+T Ethernet Box)
F93 / 10A	LAN
F94 / 2A	Compass (Quadrans)
F95 / 4A	VHF
F96 / 2A	Mantracourt T24

Please note that any nav station breaker running multiple consumers is linked to a series of fused din rail terminal blocks. These terminal blocks are located next to the DC distribution box.

A separate document will be provided by Nautor detailing the instrument power supply.

FARO INSTRUMENTATION SYSTEM

The instrumentation system on-board CS-125 is based around a FARO ATOM processor and a variety of EtherCAT terminal modules. The hardware is manufactured by Beckhoff, a leader in industrial automation systems. The EtherCAT bus provides 100 Mb/sec transfer rate between the terminals and the main processor, ensuring a real-time information exchange between the processor and the sensors.

The processor oversees the decoding of instrument data, managing math relations and outputting the data to the displays and navigation software. The system supports a very large variety of terminal modules enabling the interfacing to almost any kind of sensor.

The FARO system on CS-125 is split into two nodes (aft and forward). The aft node which also includes the main processor is located at the navigation station. The forward node is located near the mast base. The two nodes are linked together via a single CAT 6 ethernet cable.

A list of modules used can be seen below:

AFT NODE:

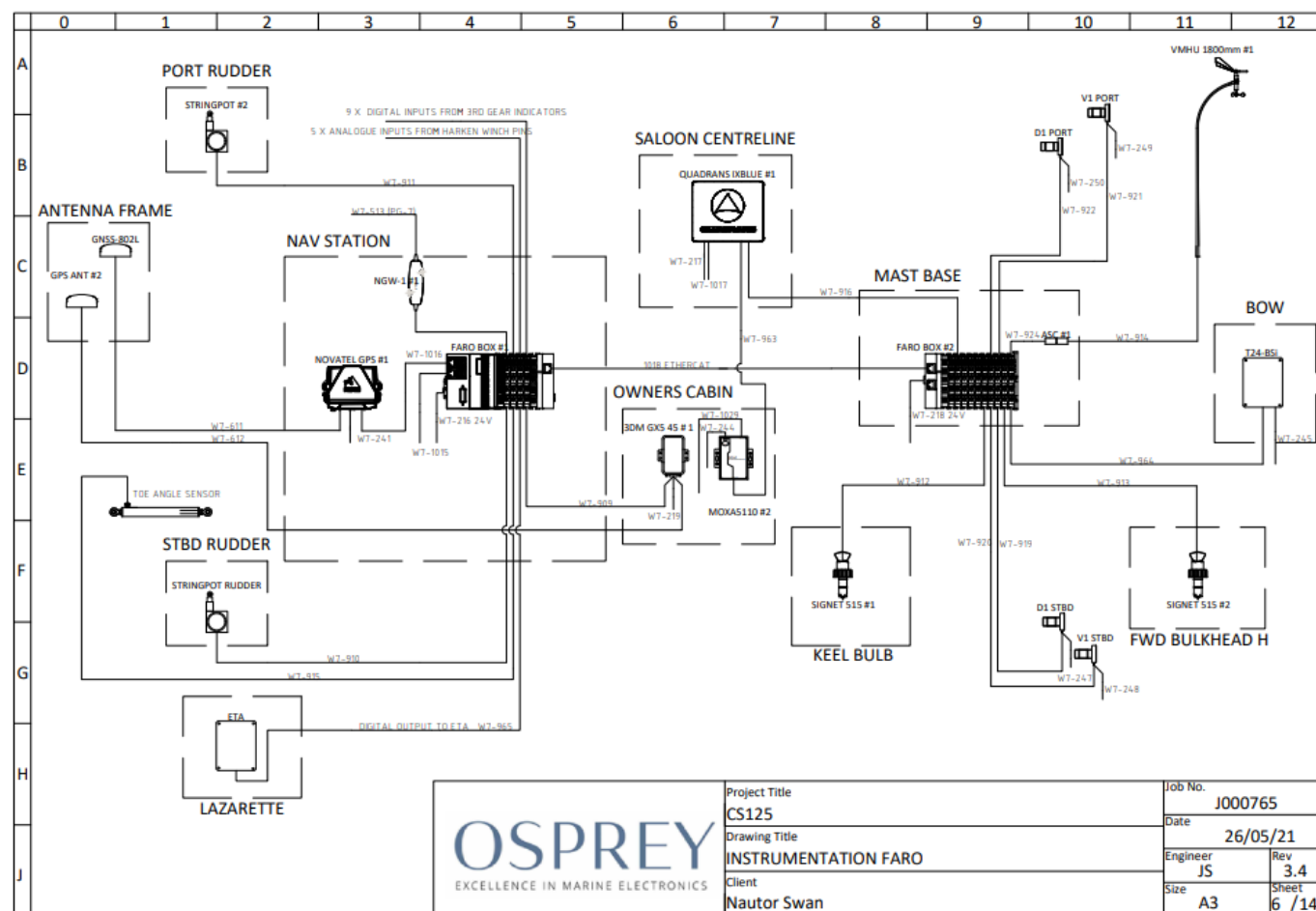
- FARO Atom Processor (2 x RJ 45, 10/100/1000 Mbit/s, DVI-D, 4 x USB 2.0, 1 x RS232/RS422/RS485, 2 x RS232)
- FARO POWER | 2 x 12V, 2 x 5V - 4 x 1A
- EL3068 | 8-channel analog input terminal 0...10 V, single-ended, 12 bits
- EL3058 | 8-channel analogue input terminal 4...20 mA, single-ended, 12 bits
- EL1809 | HD EtherCAT Terminal, 16-channel digital input 24 V DC
- EL2024 | 4-channel digital output terminal 24 V DC, 2 A
- EK1110 | EtherCAT Extension

FWD NODE:

- EK1100 | EtherCAT Coupler
- FARO POWER | 2 x 12V, 2 x 5V - 4 x 1A
- 2 x FRSGN | Signet paddlewheel sensor terminal
- EL1512 | 2-channel, up/down counter, 24 V DC, 1 kHz, 16bit
- EL6001 | Serial interface RS232
- EL6021 | Serial interface RS422/RS485
- 3 x EL6022 | 2-channel serial interface RS422/RS485, D-sub connection
- EL9011 | Bus end cover

SYSTEM DIAGRAM

The following schematic shows the layout of the FARO instrument system along with all the sensors.

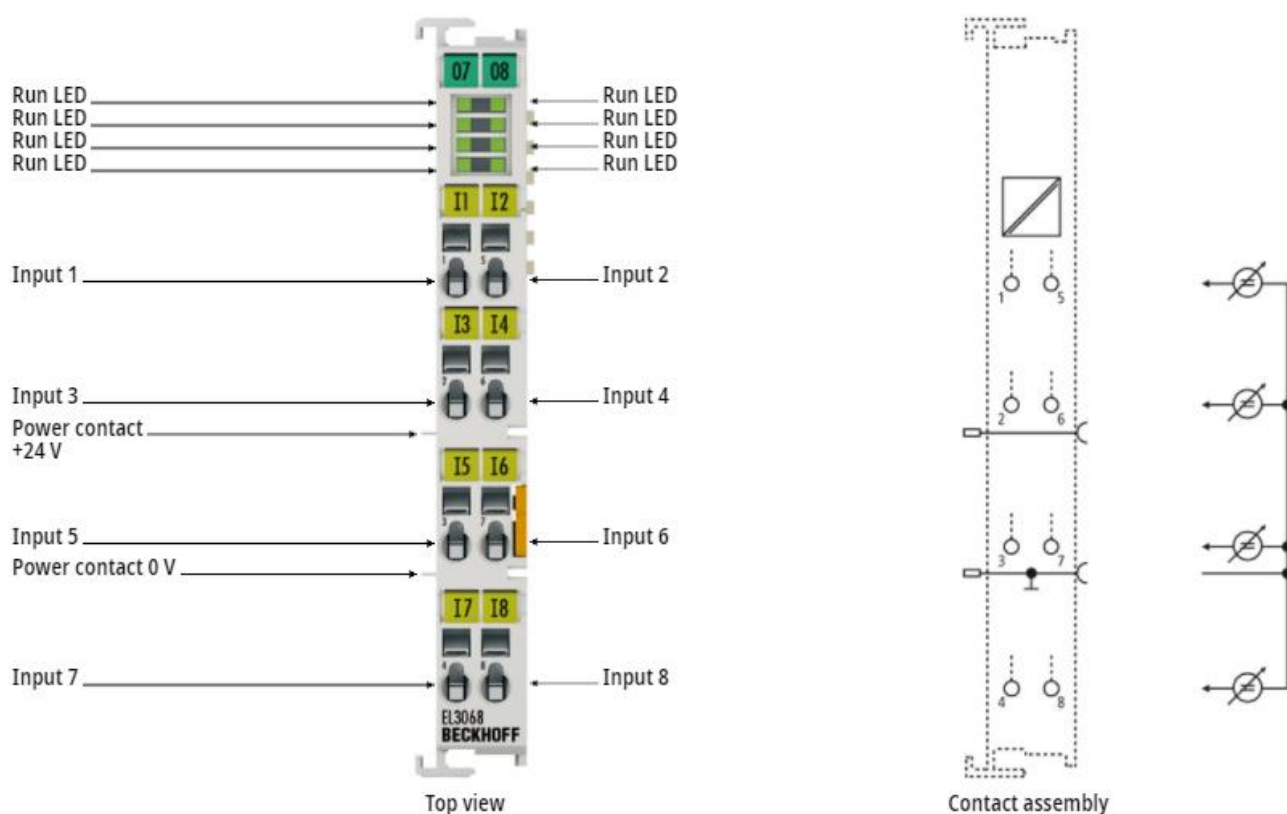


ETHERCAT TERMINALS

EL 3068

The EL3068 analog input terminal processes signals in the range between 0 and 10 V. The voltage is digitised with a resolution of 12 bits and is transmitted (electrically isolated) to the higher-level automation device.

The EL3068 EtherCAT Terminal combines eight channels in one housing. The power contacts are connected through. The 0 V power contact is used as reference ground connection for the inputs. The signal state of the EtherCAT Terminal is indicated by light emitting diodes.

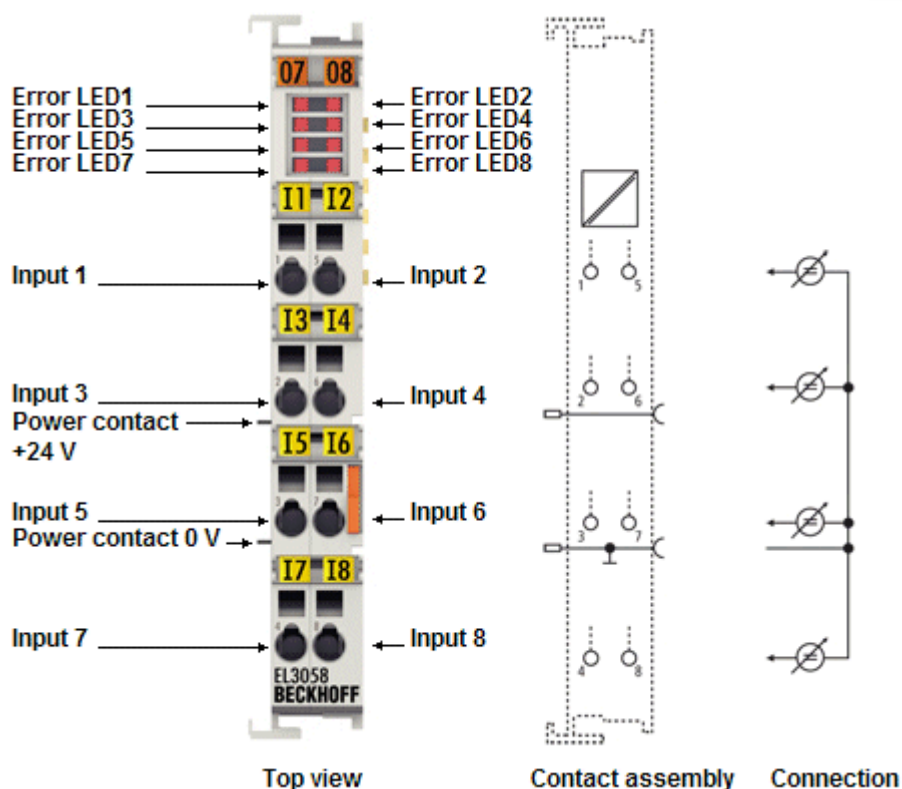


EL3068 | 8-channel analog input terminal 0...10 V, single-ended, 12 bit

EL 3058

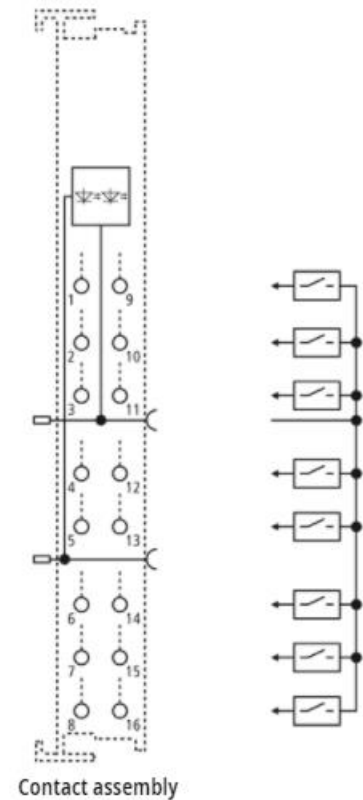
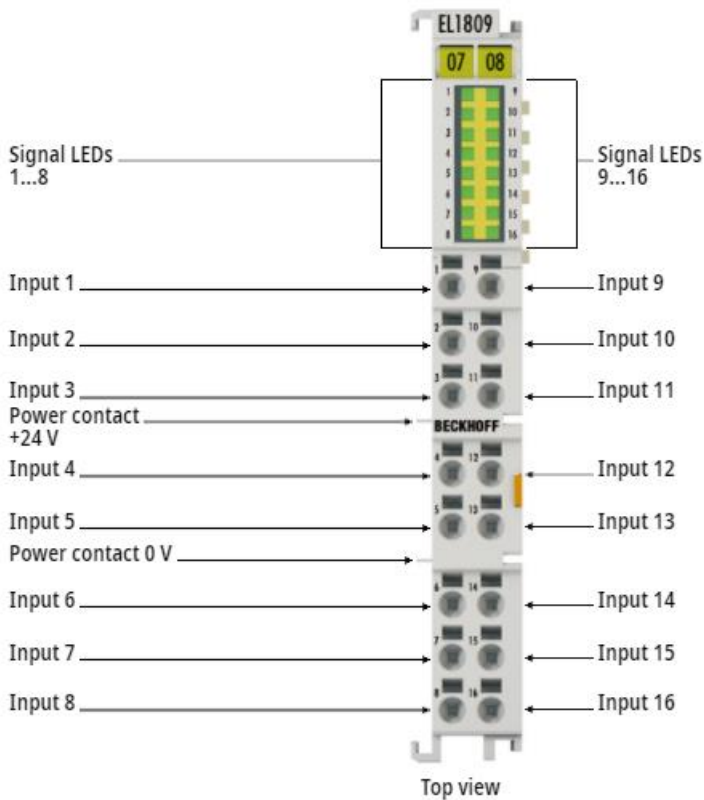
The EL3058 analogue input terminal processes signals in the range between 4 mA and 20 mA. The current is digitized to a resolution of 12 bits, and is transmitted, in an electrically isolated form, to the higher-level automation device.

The input electronics is independent of the supply voltage of the power contacts. The power contacts are connected through. The reference ground for the inputs is the 0 V power contact. The error LEDs indicate an overload condition and a broken wire. The EL3058 combines eight channels in one housing.



EL 1809

The EL1809 digital input terminal acquires the binary 24 V control signals from the process level and transmits them, in an electrically isolated form, to the higher-level automation unit. The EtherCAT Terminal has 16 channels, whose signal states are indicated by LEDs. The power contacts are connected through. With the EL1809, the reference ground for all inputs is the 0 V power contact.



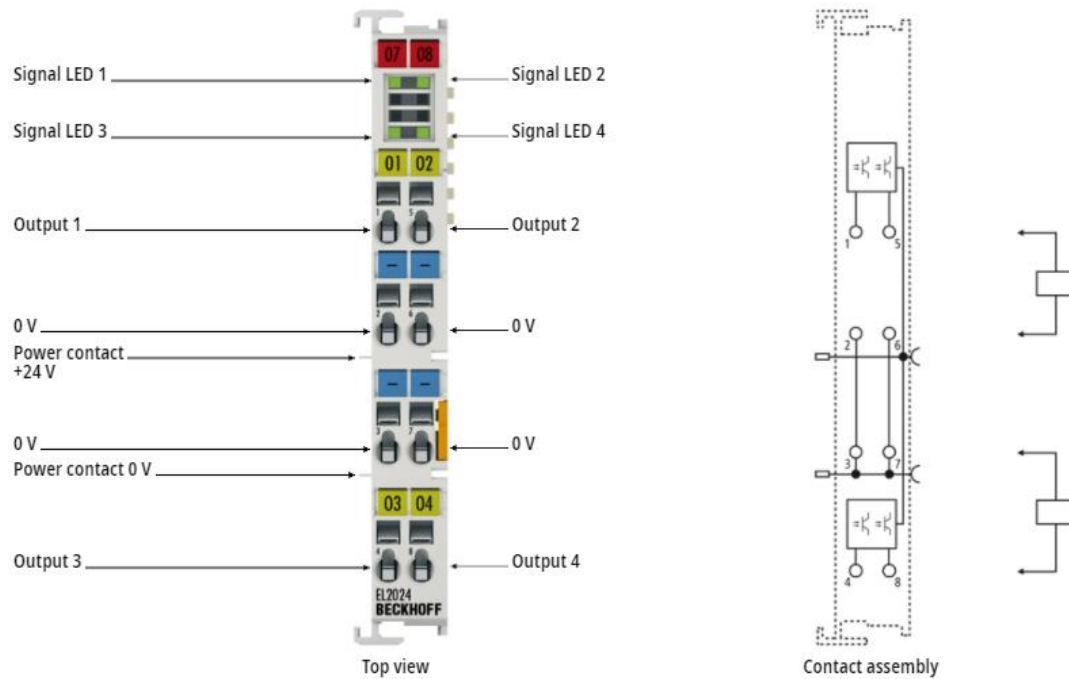
EL1809 | HD EtherCAT Terminal, 16-channel digital input 24 V DC

EL 2024

The EL2024 digital output terminal connects the binary 24 V DC control signals from the automation device on to the actuators at the process level with electrical isolation. The [EtherCAT](#) Terminal contains four channels whose signal state is indicated by LEDs. Direct connection of four 2-wire actuators is possible because the terminal has four ground connection points.

Special features:

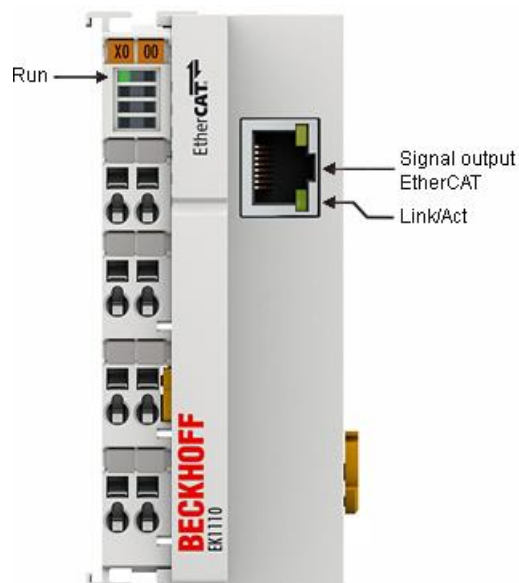
- Connection of different load types possible (ohmic, inductive, lamp load)
- Max. output current 2 A per channel
- Short-circuit-proof outputs and reverse polarity protection



EL2024 | 4-channel digital output terminal 24 V DC, 2 A

EK 1110

Like the E-bus end terminal, the EK1110 and accordingly the EK1110-0008 EtherCAT extension is connected to the end of the EtherCAT Terminal Block. The terminal offers the option of connecting an Ethernet cable with RJ 45 or rather M8 connector, thereby extending the EtherCAT strand electrically isolated by up to 100 m. In the EK1110 (EK1110-0008) terminal, the E-bus signals are converted on the fly to 100BASE-TX Ethernet signal representation. Power supply of both terminals to the electronics is via the E-bus. No parameterization or configuration tasks are required.



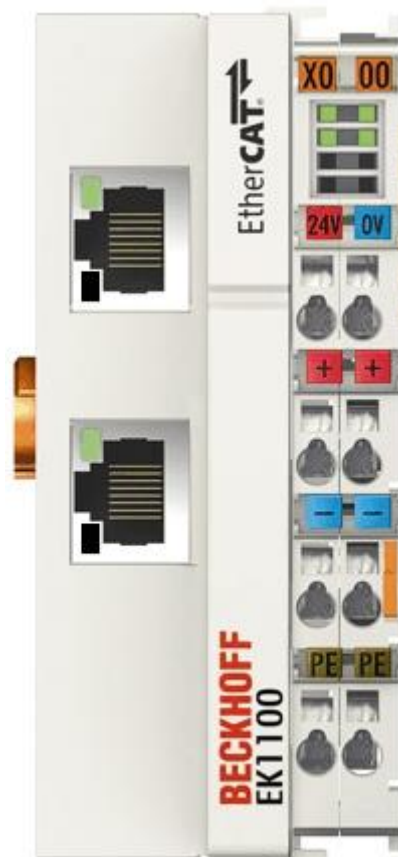
EK 1100

The EK1100 coupler connects the EtherCAT Device Protocol with the EtherCAT Terminals (ELxxxx/ESxxxx/EMxxxx). One station consists of a coupler, any number of EtherCAT Terminals and a bus end terminal, e.g. EL9011.

The coupler converts the telegrams from Ethernet 100BASE-TX to E-bus signal representation in passing with minimum latency. The coupler is connected to the network via the upper Ethernet interface. The lower RJ-45 socket may be used to connect further EtherCAT devices in the same strand.

The coupler supplies the connected terminals with the necessary E-bus current for communication. The coupler can supply a maximum of 5V/2A. Power feed terminals (e.g. EL9410) must be integrated if more current is required.

In the EtherCAT network, the EK1100 coupler can be installed anywhere in the Ethernet signal transfer section (100BASE-TX). The coupler thereby processes exclusively unaddressed MAC Broadcast telegrams of the type EtherCAT Device Protocol from the EtherCAT master. Since directed addressing via MAC Unicast or IP addressing is not used, neither a switch nor a router can be used.



FRSGN & FRPWR

Apart from the Beckhoff supplied terminal FARO has developed three custom components that can be found in this system. These are:

- FRSGN: Used to read the input from a signet paddlewheel

- FRPWR: To provide a regulated 5V and 12V supply. [2 x 5V & 2 x 12V – 4 x 1A]

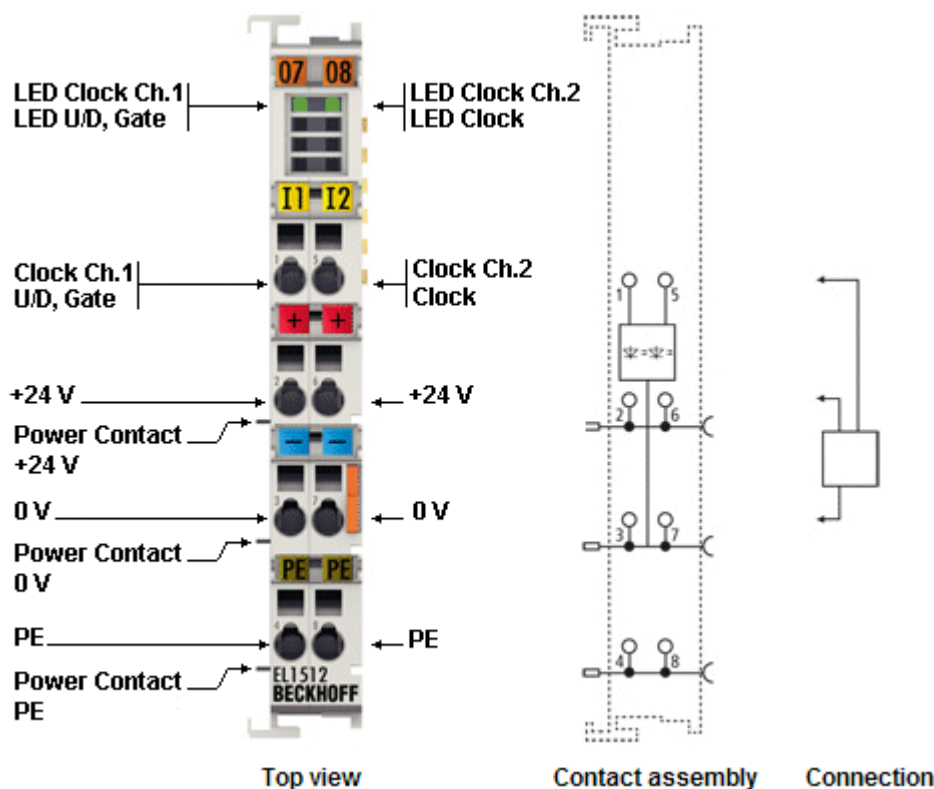
EL 1512

The up/down counter counts binary pulses, and transmits the counter state, in an electrically isolated form, to the higher-level automation device.

The EL1512 EtherCAT Terminal can alternatively be operated as:

- a single-channel counter (32 bit) that can be toggled between counting up and down via the U/D input
- a single-channel counter (32 bit) that is controlled via the gate connection
- two separate logic counters (32 bit) that can count in one direction only with the Clock 1 and Clock 2 inputs (delivery state)

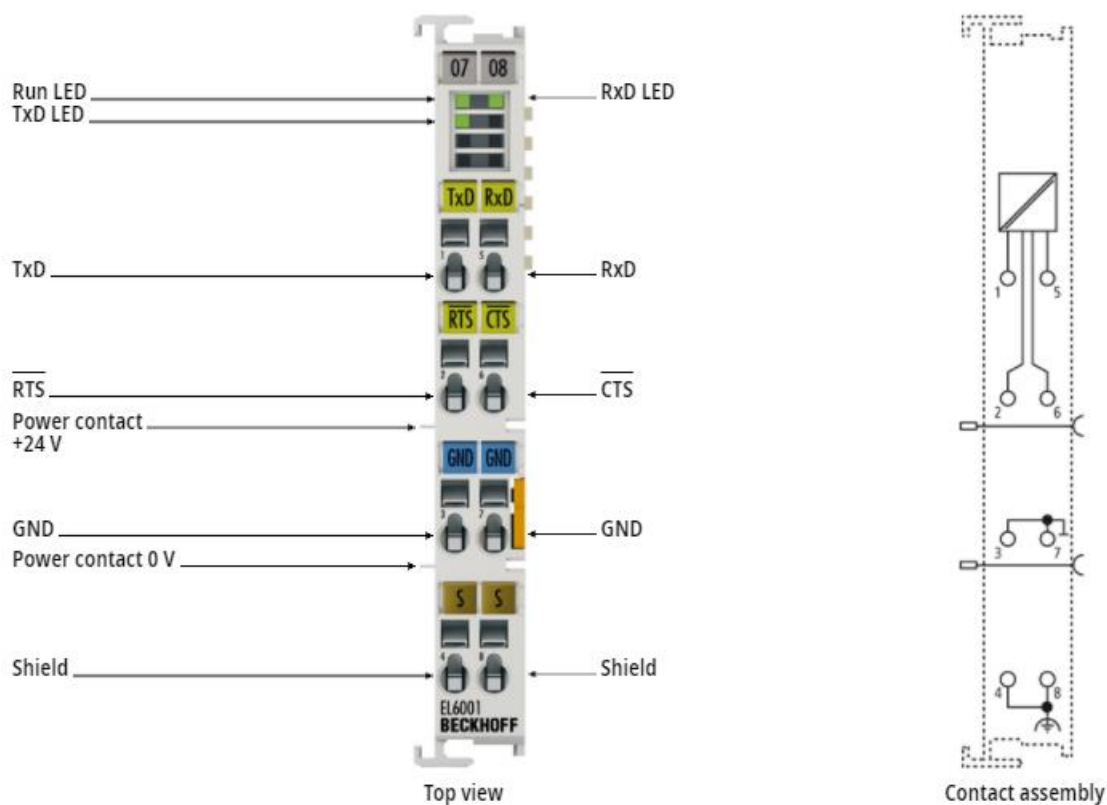
The signal state of the inputs and outputs is indicated by light emitting diodes. The EL1512 is particularly suitable for simple counting tasks such as the connection of a flow meter.



EL 6001

The EL6001 serial interface enables the connection of devices with RS232 interface and operates in conformity with the CCITT V.28/DIN 66 259-1 standards. The device connected to the EL6001 EtherCAT Terminal communicates with the automation device via the coupler. The active communication channel operates independently of the higher-level bus system in full duplex mode at up to 115.2 kbaud. The RS232 interface guarantees high immunity to interference through electrically isolated signals.

In conjunction with the TwinCAT Virtual Serial COM Driver (see TwinCAT Supplements – Communication) the EL6001 can be used as a normal Windows COM interface.

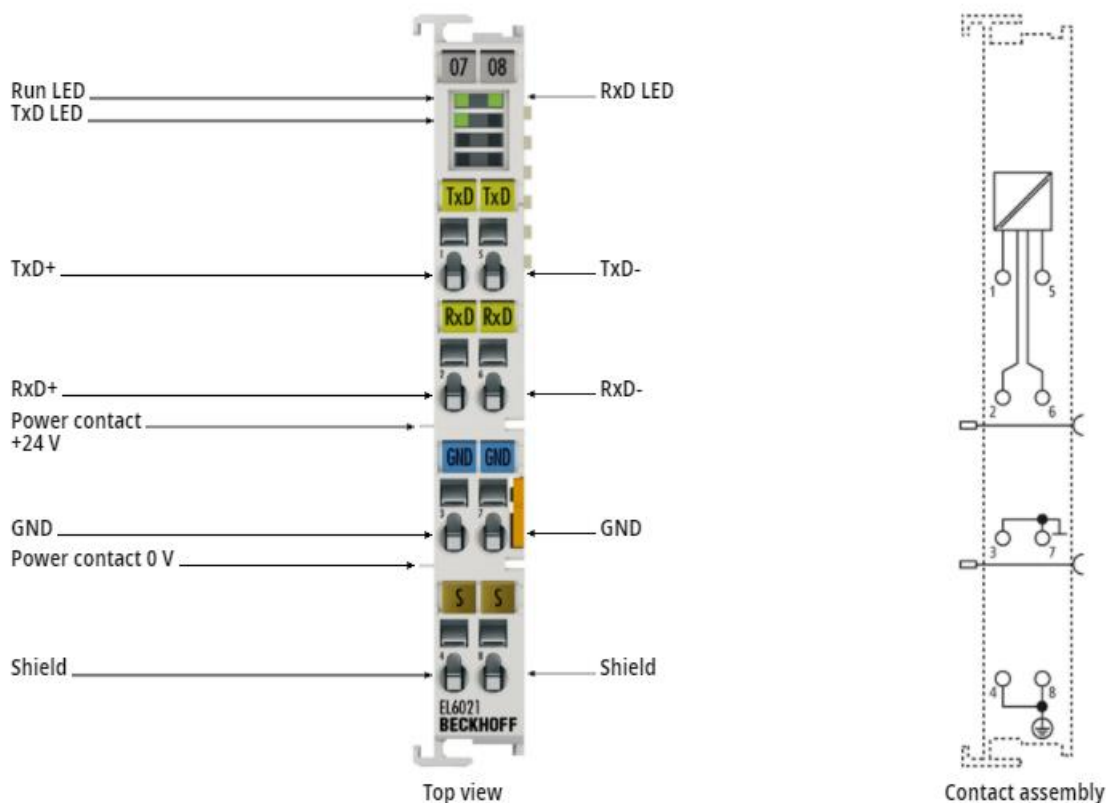


EL6001 | Serial interface RS232

EK 6021

The EL6021 serial interface enables the connection of devices with RS422 or RS485 interface. These devices communicate with the automation device via the coupler. The active communication channel operates independently of the higher-level bus system in full or half duplex mode at up to 115.2 kbaud. The transmission of differential signals according to RS422 guarantees high immunity to interference through electrically isolated signals.

In conjunction with the TwinCAT Virtual Serial COM Driver (see TwinCAT Supplements – Communication) the EL6021 can be used as a normal Windows COM interface.



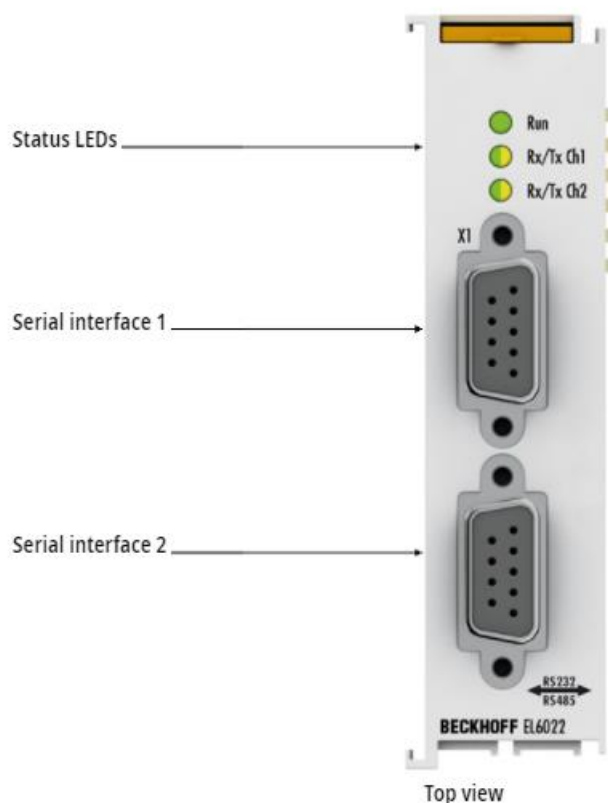
EL6021 | Serial interface RS422/RS485

EK 6022

The EL6022 serial interface enables the connection of devices with two RS422/RS485 interfaces. The devices connected to the EL6022 EtherCAT Terminal communicate with the automation device via the Coupler. The active communication channel operates independently of the higher-level EtherCAT system in full duplex mode with 300 baud up to 115.2 kbaud.

The RS422/RS485 interfaces guarantee high interference immunity through electrically isolated signals. The EL6022 can provide 2 x 5 V/20 mA out of the E-bus supply (electrically isolated, short-circuit proof) for the supply of external devices.

In conjunction with the TwinCAT Virtual Serial COM Driver, the EL6022 can be used as a normal Windows COM interface.



EL6022 | 2-channel serial interface RS422/RS485, D-sub connection

EL 9011

Each bus station must be terminated on the right-hand side with an end cap. The EL9011 covers the E-bus contacts and is colour-matched to the EL terminal series.

FARO MODULE PIN-OUT

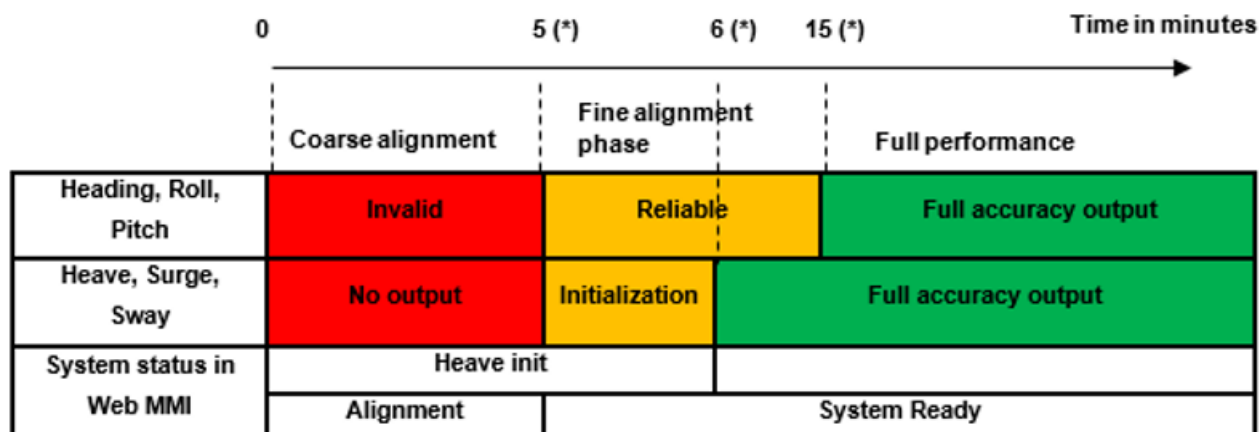
Accompanied with document is a detailed excel sheet with both FARO node pinouts.

SENSORS

QUADRANS IXBLUE

The primary heading sensor is an IXblue Quadrans Fibre-Optic-Gyro. This is an inertial navigation system that outputs heading, heel and trim with rates to the FARO system.

As this compass is not magnetic it requires at least 5 minutes of coarse alignment time upon boot before outputting any data. After the fine alignment phase, which can take up to 15 minutes, the FOG outputs fully accurate data.



(*) Fixed time (**) Variable time

- Stastic vehicle, no move
- Vehicle can freely move.
- Full system performance

Figure 1 – QUADRANS Starting Sequence (at powering on or software restart)

At the end of the fine alignment process, QUADRANS is ready for navigation with optimal performance. During navigation, QUADRANS uses its own inertial sensors together with GPS, when it is available, to provide optimal estimates of position, speed, attitude and heading. When no external sensor is available, QUADRANS automatically switches to “pure inertial mode” and keeps providing position, speed, attitude and heading estimates.

It is individually powered via a 24V DC supply and its interfaced to the instrument system via a serial connection to a EL6001 module on the forward FARO node.

The FARO reads the serial information on COM6 at a baud rate of 38400 using the INDYN protocol. GGA position information is also sent to the Quadrans on the same com port.

The ethernet connection is also utilised and connected to the mast base SD8 switch. This is primarily used for diagnostics and configuration.

The Quadrans webserver can be accesses on the following IP address: 192.168.10.12

LORD3DM-GX5-45 GNSS/INS

The MicroStrain Sensing 3DM-GX5-45 is an all-in-one navigation solution featuring high-performance, integrated multi-constellation GNSS receiver, accelerometer, and gyro.



LORD 3DM Highlights:

- High-performance integrated multi-constellation GNSS receiver and advanced MEMS sensor technology provide direct inertial measurements, and computed position, velocity, and attitude outputs in a small package
- Triaxial accelerometer, gyroscope, magnetometer, temperature sensors, and a pressure altimeter achieve the optimal combination of measurement qualities
- Dual on-board processors run a new Auto-Adaptive Extended Kalman Filter (EKF) for outstanding dynamic position, velocity, and attitude estimates

The MicroStrain 3DM is used as the secondary GPS, Heading, Pitch, Roll and Gyro sensor. It is interfaced with the FARO via an RS232 serial connection to the main FARO processor. The assigned com port on the FARO is COM3 with a baud rate of 115200. The protocol used is the Sp_3DM_MIP.

The sensor is located under the aft central floorboard in the owner's cabin. Its connected to a Tallysman TW3710 GPS antenna mounted on the antenna frame.

The sensor can be configured via the Sensor Connect software or via MIP Monitor.

NOVATEL PWRPAK 7

The primary GPS is a Novatel PwrPak 7 unit.

The PwrPak7 is a compact enclosure that delivers scalable Global Navigation Satellite System (GNSS) with internal storage and INS options. It's capable of tracking all present and upcoming GNSS constellations and satellite signals. It also offers optional integrated INS support for continuous position, velocity, and attitude through short periods of GNSS outage.

With its powerful GNSS engine, 16 GB of internal storage and enhanced connection options including serial, USB and Ethernet, the PwrPak7 is ideal for many applications



The main unit is mounted at the nav station with a VEXXIS GNSS-802L antenna mounted on the antenna frame. The PwrPak 7 is configured to communicate with the FARO system via UDP protocol. The IP address is 192.168.10.11 and transmits data on port 3002.

The device has been configured via the Novatel Connect app to transmit the following NMEA sentences:

- GGA: Global Positioning System Fix Data (Includes but not limited to - Time, Latitude, Longitude, Fix Quality, Number of Satellites and HDOP)
- RMC: Recommended minimum specific GPS/Transit data (Includes – UTC of position fix, Latitude, Longitude, SOG, Course Made Good in degrees True, UT date and Magnetic Variation)
- VTG: Track Made Good and Ground Speed (includes – True Track Made Good, Magnetic Track Made Good and SOG in Knots)

ZG100 GPS ANTENNA

The tertiary GPS unit is a dual function B&G ZG100 GPS antenna and compass. It is a NMEA 2000 device connected to the main instrument CAN bus and outputs GPS data at 10Hz. It also provides heading, heel and trim data.



The sensor is mounted on the antenna frame. Data coming from this sensor, as per all NMEA200 devices, is interfaced with the FARO via an Actisense NMEA2000 to USB converter (NGT-1-USB).

In the unlikely event of a FARO failure this would act as the backup GPS. Sensor data could be read but the A+T Processor and shown on selected A+T displays.

RUDDER REFERENCE UNITS

Each rudder has a TE cable actuated position stringpot attached to it. The part number for the stringpot is SPD-25-3. It has a 25" inch range and a 4-20mA output. These are IP 67 rated.

The outputs of these sensors are wired into a EI3058 module at the navigation station FARO enclosure.



SPD-25-3

There is an additional Simrad RF300 rudder reference unit connected to the starboard rudder for use with the removable autopilot. The RF300 is wired directly into the NAC-3 autopilot computer via a Deutsch ASC connector for quick and easy removal.



Simrad RF300

SIGNET 515 PADDLEWHEEL'S

Both speed sensors are Signet 515 models. The primary sensor is mounted on the keel bulb with the backup sensor immediately forward of bulkhead H on centreline. Both sensors are connected to the fwd FARO module and have connectors for quick removal.



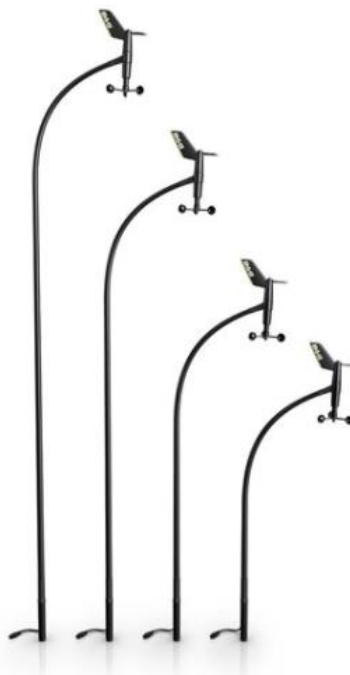
B&G MASTHEAD UNITS

There are four masthead units (MHU's) in total. Two 1.8m ocean spec WS740S models and two 1.1m ocean spec WS720S models. A 1.8m and 1.1m MHU will be mounted at the top of the rig with the rest kept as spare.

These are the latest MHU's by B&G and offer increased aerodynamics, stability, and accuracy. The output of these new MHU's is NMEA0183 data. The main 1.8m wand is wired into a El6021 serial module on the fwd FARO node. The backup 1.1m MHU is wired into an interface, converting 0183 data into NMEA2000, and then onto the main instrument CAN bus.

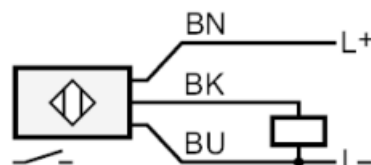
In the unlikely event of a FARO failure, data from the backup wand would be available to the A+T processor for processing and display.

The cabling running through the rig is a custom type 44 4C cable with blue ASC connectors at the mast base and masthead. Each wand connector is compatible with either of the two wind cables.



HARKEN IE5369 INDUCTIVE SENSORS

Seven out of the nine winches come with IE5369 inductive sensors. These are used to indicate and log when the winches are in the 3rd gear. The sensors are part of Harkens supply with Osprey interfacing them to the FARO system.



Core colours :
BK = black
BN = brown
BU = blue

IE5369 Sensor Wiring

All seven sensors are powered from within the nav station FARO enclosure and the signals are wired into the EL1809 digital input module. Please refer to schematics and pinout sheet for more info.

HARKEN WINCH PINS & AMPLIFIERS

The primary, runner and main sheet winches are all fitted with Harken winch load pins. These are Harken supplied along with the amplifiers and Osprey have interface them to the FARO system. The loadcell amplifiers are mounted below decks at the base of each winch and are powered from within the nav station FARO enclosure. The output of each amplifier is wired into a separate input on the EL3058 module at the nav station FARO aft node. Please refer to schematics and pinout sheet for more info.

MANTRACOURT DCELLS & T24-SO

The rig load sensing package is supplied by Future Fibres. (Please note this is separate to hydraulic ram pressure data which is part of the Cariboni package.) This package includes four wired sensors for the shrouds (V1 Port, V1 Stbd, D1 Port & D1 Stbd) and 4 wireless load sensors for the J0, J2, J2 strop and J4.

DCells

Future Fibres have used Mantracourt DCell converters with the V1 and D1 strain gauges to convert the output of the strain gauges into a calibrated serial RS485 ASCII string which can be interfaced to the FARO. The outputs of each V1 and D1 are wired into individual RS485 serial ports on the fwd FARO node. The FARO serial communications protocol used is SpSerialValue.

Load Sensor	FARO Manager COM Port & Baud Rate
V1 Stbd	COM 8, 115200

D1 Stbd	COM 9, 115200
V1 Port	COM 10, 115200
D1 Port	COM 11, 115200

Each DCell has been configured with a STN number of 998 for continuous ASCII output. Software used to configure the DCells is Instrument Explorer by Mantracourt Electronics.

On the FARO side the serial output of the DCells have been wired into MOXA DB9F to TB adapters. This has been done so that they can easily be disconnected from the FARO and linked up to a computer for diagnostics and configuration. A MOXA UPort 1130 Serial to USB converter has been supplied to help with this.

The V1's and D1's are all powered from the mast base instrument panel. The supply is a 12V DC supply from the mast base FARO power module.

T24-SO

The T24-SO is used in conjunction with the wireless transmitters. It is a wireless receiver that generates a serial RS485 user designed ASCII output. It can include data from up to 8 x radio telemetry T24 transmitters.

In this case the data from the four transmitters is combined into a single ASCII string. The output of the T24-SO is wired into a EL6022 module at the mast base fwd FARO node. In FARO Manager it appears as COM 12 with a baud rate of 115200. The SpNmeaIn FARO serial protocol has been used.

The T24-SO is mounted under the deck in the forward bow compartment. This has been done so that its as close to the transmitters as possible.

For configuration, calibration and testing the T24 Toolkit software can be used.

A T24-BSU (Wireless Telemetry USB Base Station) has also been supplied for easier configuration of the T24 telemetry modules.

The T24-So is powered by a 24V DC supply from the mast base panel DC distribution box.

A+T CPU & DISPLAYS

Along with the FARO system there is an A+T ATP1 CPU installed at the navigation station. The primary role of the A+T CPU is to act as a server for the MFD and BFD displays. The data to be displayed is sent via a UDP stream from the FARO to the ATP1. The ATP1 processes this data stream and sends it out to the displays in a proprietary format.

The A+T display network has been split into two networks. The aft displays comprising of 13 MFD's and the fwd network comprising of 3 BFD's located on the mast display bracket.

Apart from acting as a display driver the ATP1 CPU also handles the MOB button inputs and drives the internal relay that activates and launches the Jon Buoy.

The ATP1 is also connected to the NMEA2000 network so in the unlikely event of a FARO failure the ATP1 would still be able to process any data coming from NMEA2000 sensors.

The IP address of the ATP1 is **192.168.10.54**

ATP1 UDP network connection setting to receive FARO data can be seen below:

Network I/O

Manage data input and output over ethernet for connection with third party software packages.

Output Frequency 1 Hz

Network Connection 1

Connection Type: UDP Input

Adapter Address: 192.168.10.54

Connection Address:

Format: Faro

Enable: ☒

Broadcast: ☒

Port: 5005

+

OK Cancel Apply

The received FARO data can be seen by selecting Other Data>Faro in the menu bar of the ATP1 webserver.

All MFD pages have to be configured locally at each display. The BFD's however can be configured from the ATP1 webserver by going to Display>Pages and then selecting the active display.

Please refer to the schematics for the exact display layout and wiring.

MOB

There are two MOB buttons in total. One per helm. When an MOB button is pressed two MFD displays have been configured to display distance and bearing to MOB as well as the time since the MOB was activated. These displays are the MFD's at the helm consoles.

In addition to the above an internal CPU relay is activated which allows for the Jon Buoy to be launched if the Jon Buoy DC breaker is on. The Jon Buoy 24V breaker is located at the nav station dc distribution box. The output is fused at 2A to protect the internal ATP1 CPU relay which is rated at 2A.

CARIBONI & PIXEL SUR MER INTERFACING

The FARO system is interfaced to both the Cariboni hydraulic PLC and the Pixel Sur Mer keel and daggerboard monitoring system.

Interfacing is done over the network via UDP. In FARO these are labelled in network devices as ExocetRead and PLCClient. Port numbers are 50008 and 50107 respectively.

CARIBONI PLC DATA

Data coming from the Cariboni PLC to the FARO contains hydraulic ram pressure data, cylinder ram position data in a percentage, engine data, ballast tank data in percentage and Cariboni oil tank temp.

Some of this data, primarily ram pressure data, is converted to tonnes and corrected/calibrated within the FARO.

This data can be logged within the FARO and alarms can be raised if required.

PIXEL SUR MER

The Exocet Blue processor sends the FARO CPU the following data:

- Fiber optic sensor data for keel and daggerboard

Raw data isn't sent so that to not overload the communication channel. Instead, for each sensor, the Exocet Blue exports the maximum data value over a 1 second time frame.

- Estimated daggerboard loads

Derived from ultrasonic sensor data and calculations provided by the design team (JYD).

The communication protocol is a UDP "NMEA-like" ASCII text protocol.

Data is also sent from the FARO to the Exocet Blue. For this we are utilising the already existing UDP stream that the FARO uses to send variable names and data to the ATP1 CPU.

Further details on communication protocols, port numbers, lists of variables etc can be found within FARO manager.

ALARMS

Within FARO manager it is possible to set alarm limits for any variable within the system. If an alarm is triggered it will be visible on the FARO manager page on the main nav computer. A digital output module has also been added to the aft FARO unit so that an alarm signal can be sent to the ETA system.

VHF & AIS

Installed are a fixed VHF unit and AIS transponder.

ONBOARD VHF

A B&G V90S VHF black box VHF unit is installed at the mast base panel. This unit has two wired handsets. One located at the nav station and one at the stbd helm. Along with the handsets there are also two external speakers installed.



The masthead VHF antenna is a Scan VHF-33-1 and dedicated only to the VHF. The coax cable running from the black box to the masthead antenna is LMR600-Lightweight.

Given the length of run from the black box to the handsets a higher spec, higher gauge handset cable has been used. The exact cable spec is: Alpha Wire B955084 GE321 8-core 20AWG tinned shielded cable.

The V90S is also an AIS receiver. The 0183 output of the VHF has been wired into a serial device server so that the AIS data is available over the ethernet network.

In the unlikely event that the primary AIS unit fails the vessels will still be able to receive AIS data via the V90S and its masthead antenna.

An emergency VHF antenna has also mounted on the top of the radar post on the starboard aft side of the vessel. The emergency VHF antenna cable is LMR400-Ultraflex and the antenna is a UltraWhip. To change from the masthead antenna to the emergency antenna someone would have to physically connect the spare cable to the V90S black box at the mast base.

AIS TRANSPONDER & AIS ALARM BOX

ATB1

The vessel is equipped with a Class B AIS transponder from Ocean Signal. It is a ATB1 Class B AIS Transceiver incorporating the superior SOTDMA access scheme.

Using the superior SOTDMA access scheme sets the ATB1 apart from the CSTDMA class B products available. This Ocean Signal Class B unit has the same AIS transmission priority as a class A unit, giving the added assurance that there will be no loss or delay of transmission, even in high traffic areas.

The ATB1 also has a faster reporting rate and higher output power than CSTDMA class B units. It sends AIS transmissions every 5 seconds instead of the maximum two transmissions per minute and the 5W output power instead of 2W allows your transmissions to reach further.

The ATB1 has its own dedicated antenna which is mounted on the top of the antenna frame. The antenna is a Scan VHF-33. The GPS antenna for the ATB1 is also mounted on the antenna frame and is part of the Taoglas 5-in-1 combined antenna.

The ATB1 is connected to the NMEA2000 network. The NMEA0183 output is wired into the AIS Alarm box at the navigation station.



ATB1 AIS

AIS Alarm Box

The **AIS alarm box** is designed to ensure that the moment any AIS MOB or AIS SART signal is received an 87dB audible alarm is sounded alerting to a possible MOB. It is interfaced with the AIS via NMEA0183-HS (38400 baud rate).

It can be programmed by the user with up to 30 individual MMSI numbers and multiple tones to identify the source of the AIS signal (MOB or SART). During the commissioning period all personal AIS unit MMSI numbers were programmed into the AIS Alarm unit.

The NMEA0183 output of the AIS Alarm Box is wired into a Serial Device Server at the nav station so that AIS data is accessible over ethernet to the nav PC and relevant navigational software.



AIS Alarm Box

VESSEL RADIO INFORMATION

The following details have been programmed into the AIS & VHF

Vessel Name: Skorprios
Call Sign: ZGOP9
MMSI: 319214500

B&G HALO RADAR

A Halo B&G radar is installed on a radar post on the stbd aft side. This is connected to the LAN and currently can only be used with the Expedition navigation software installed on the main navigation PC (PC1)

ANTENNAS

PRIMARY NOVATEL GPS ANTENNA – GNSS-303L

The GNSS-802L is the main GPS antenna for the Novatel PwrPak7 GPS unit. It maintains a low profile while achieving both high peak zenith gain and low gain roll-off from zenith to horizon, without sacrificing tracking performance. This technology significantly enhances the low-elevation angle tracking capabilities, extending operation to the entire GNSS constellation. Furthermore, the GNSS-802L, with L-Band, can achieve a greater phase centre stability, which translates into an improved carrier phase measurement and a better RTK solution.

Some of its features are:

- Supports dual-frequency GPS and GLONASS signals
- L-Band signal reception, supporting correction services such as TerraStar
- Multi-point antenna feed provides stable phase centre and enhanced multipath rejection
- Radiation pattern optimization technology yields exceptional low elevation satellite tracking

- Provides exceptional tracking performance previously unachievable in a small form factor
- Hermetically sealed enclosure to endure the toughest environments

It's mounted on the antenna frame.



GNSS-303L Antenna

STORM MA450 ANTENNA

The Storm MA450 antenna is a 5in1 low profile, heavy-duty, fully IP67 waterproof external antenna for use in worldwide telematics and IoT applications which require best in class LTE, GNSS, and Wi-Fi performance. It delivers powerful worldwide 4G LTE MIMO antenna technology, plus GPS-GLONASS-GALILEO-BeiDou for next-generation location accuracy. The antenna also covers legacy 2G and 3G bands for devices that fallback where 4G is unavailable and the dual-band MIMO Wi-Fi antennas enable high throughput Wi-Fi speeds.

The MA450 antenna features:

- 2 x 4G/3G/2G MIMO Antenna
- 2 x Wi-Fi 2.4GHz/5.8GHz MIMO Antenna
- 1 x GPS/GLONASS/GALILEO/BEIDOU L1 Antenna

Two 4G and a Wi-Fi antenna are used by the HD2 router. The GPS antennas is used by the AIS unit.

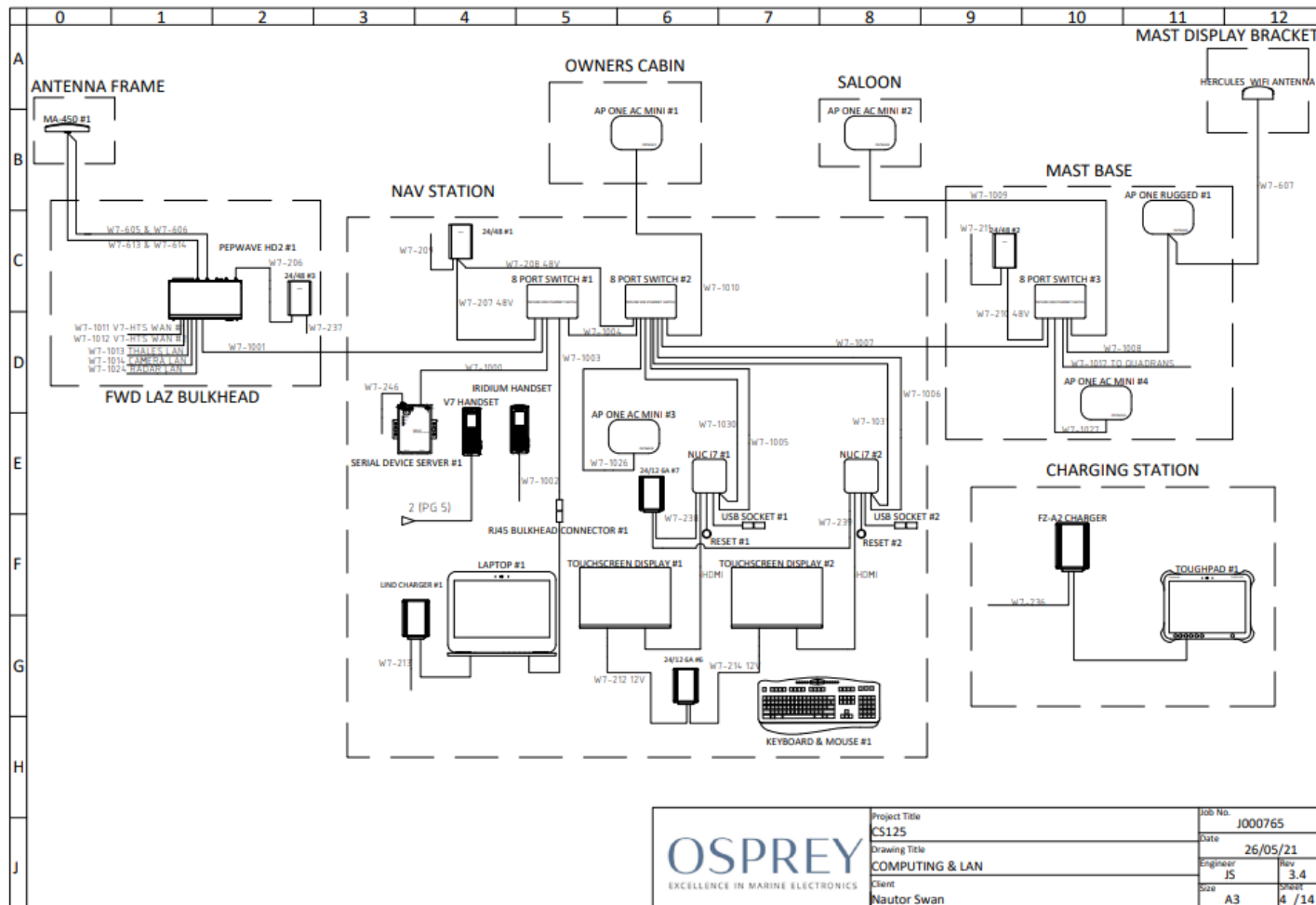
This is mounted on the antenna frame.

ON DECK MAST DISPLAY -WIFI ANTENNA

The Wi-Fi antenna mounted on the top of the mast bracket is a Taoglas Hercules WS.01 dual band (2.4GHz/5,2GHz) IP67 rated antenna.

COMPUTER NETWORK OVERVIEW

Below is a logical layout detailing the core network onboard:



COMPUTER NETWORK

CS-125 features a robust network that runs from the Mast Base, through the Nav Station and into the Lazarette. The network is built on Peplink Industrial switches that feature 2 gigabit SFP ports and 8 10/100-megabit ports of which they can all provide 802.3af Power over Ethernet.

The network is managed by the Pepwave HD2 which acts as the router (or Gateway in and out of the network), DNS server and DHCP server issuing IP addresses to joining clients on the network. The HD2 also acts as an uplink controller allowing the boat to choose how it obtains an internet connection. Currently the only connection available is via 4G.

Peplink SD 8 switches just like the HD2 can be configured via their web page. The web configuration page for each switch can be reached by typing in the IP address of the switch into a web browser.

Peplink Equipment Location	IP address
HD2 - Lazarette	192.168.2.1
Switch #1 – Nav Station	192.168.2.51
Switch #2 – Nav Station	192.168.2.50
Switch #3 – Mast Base	192.168.2.52
AP One - Nav Station	192.168.2.55
AP One Rugged – Mast Base	192.168.2.56
AP One – Owner Cabin	192.168.2.57
AP One - Saloon	192.168.2.58
AP One – Mast Base	192.168.2.59

The login for the switches and HD2 is admin / OspHd220_!

ROUTER

The vessel is fitted with a Pepwave HD2 cellular router in the lazarette compartment. This device manages the boats WAN (Wide Area Connection) and LAN (Local Area Connection). It is managed through a web interface and can be reached at 192.168.2.1

It features a DHCP server to issue IP addresses to clients on the network. On the LAN interface multiple VLAN's have been configured.

VLAN'S

Various VLAN's have been configured in order to provide easier management and ensure greater network flexibility and performance.

These are listed in the table below:

LAN Name	VLAN	Apply to	IP Settings for Switch	Default IP address
Management Port	None	N/A	192.168.1.254/24	
Untagged LAN	None	None	N/A	
Default VLAN	1	All SD Switch	None	192.168.2.0/24
VIP_Wifi	11	All Balance/MAX and SD Switch	None	10.1.1.1
Captain-wifi	13	All Balance/MAX and SD Switch	None	10.1.3.1
Crew-Wifi	14	All Balance/MAX and SD Switch	None	10.1.4.1
Faro	32	All Balance/MAX and SD Switch	None	192.168.10.1
Cameras	33	All Balance/MAX and SD Switch	None	10.3.3.1
Iridium	99	All SD Switch	None	192.168.55.1
ETA	310	All Balance/MAX and SD Switch	None	10.3.10.1

WAN SELECTION

The WAN interface allows the ability to choose between Cellular, Wi-Fi WAN and satellite connections as a source of internet. This is done from the main dashboard of the HD2 web interface.

WAN Connection Status

Priority 1 (Highest)

Cellular 2 Connected to DNA 3G [Details](#)

Priority 2

Drag desired (Priority 2) connections here

Priority 3

1 WAN 1 - V7 Unlimi... No Cable Detected [Details](#)

2 WAN 2 - V7 HS No Cable Detected [Details](#)

Priority 4 (Lowest)

Drag desired (Priority 4) connections here

Disabled

Wi-Fi WAN Disabled [Details](#)

Cellular 1 Disabled [Details](#)

CELLULAR

The HD2 has 2 SIM card slots. This allows you to switch between data tariffs or when you have used your quota on one, move to the other. It is important to remember that the HD2 only has 1 modem so can only manage one SIM/Cellular connection at any given time.

APN information is set to automatic.

Remember to enable 'Data Roaming' if you have a SIM that is valid for roaming between countries.

The signal threshold can be adjusted should the need to switch to or away from cellular need to be adjusted if the signal is too weak to an acceptable connection.

WAN 1 - V7 HTS UNLIMITED

WAN #1 of the HD2 is connected to the KVH V7 HTS unlimited channel.

WAN 2 – V7 HTS HS

WAN #2 of the HD2 is connected to the KVH V7 HTS high speed channel.

WIFI WAN

Wi-Fi WAN can be used to connect to a marina/yard Wi-Fi but also to the Iridium Thales SSID.

WIRELESS NETWORK

An Enterprise level wireless network is installed utilizing Pepwave equipment. This is capable of broadcasting multiple Wi-Fi SSID and clients. This hardware also allows 'handoff' or roaming allowing a user to seamlessly move from one access point to the next. This infrastructure is made up of six access points. These are the AP One Rugged at the mast base, the HD2 in the lazarette and the AP One Enterprise access points in the nav station, owners' cabin, saloon, and mast base. The HD2 and the AP One Rugged have on deck Wi-Fi antennas. One is located on the aft antenna frame and the other on the mast display bracket.

The HD2 acts as the access point controller and is set to broadcast three different SSID's. The details are as follows:

SSID	Password	SSID Availability	VLAN
CS125_Wifi	cs125_11	All Devices	VIP_Wifi (VLAN ID:11)
CS125_Crew	cs125_14	All Devices except Owner Cabin AP	Crew_Wifi (VLAN ID:14)
CS125_Tech	cs125_32	All Devices except Owner Cabin AP	Faro (VLAN ID:32)

- CS125_Wifi: Intended for the owner/VIPs. Clients will NOT have access to any systems on the yacht. They can access the internet via all options including the High Speed VSAT. Traffic on this network has priority.

- CS125_Crew: Intended for general crew, maybe race crew. Clients will NOT have access to any systems on the yacht. They can only access the internet via the cellular connection.
- CS125_Tech: Intended for technicians. Clients will have access to all systems on the yacht. They can also access the internet via all options apart from the High Speed VSAT

Local Access / User groups (aboard the vessel)	SSID to use	Internet Access Permission			
		VSAT Highspeed	VSAT Unlimited	Iridium Certus	Cellular
Owner/VIP	CS125_Wifi	✓	✓	✓	✓
Technical Crew - Captain, Engineer, Navigator	CS125_Tech		✓	✓	✓
General crew/race crew	CS125_Crew				✓

SERIAL-TO-ETHERNET SERVER

A Moxa NPort 5250A is installed in the navigation station to convert serial data into an ethernet accessible method. This device is connected to the AIS transponder to receive AIS data.

A secondary Moxa NPort (5150A) is also installed at the mast base electronics panel. This receives AIS data from the V90S VHF. Note the V90S is only an AIS receiver. This should be used as a back up in the event of primary AIS failure.

The MOXA NPort's can be controlled/configured via NPort Windows Driver Manager.



COM PORTS

There are several COM ports available to the computer for access to different parts of the system. Using the 'NPort Windows Driver Manager' Software the computer can emulate the ports, via the Ethernet connection, as if they are physically connected to the computer.

Computer Port	NPort Port	Purpose	IP Address	Baud Rate
COM 1	Port 1	Main AIS Data	192.168.10.23	38400
COM 2	Port 2	Not Used	192.168.10.23	-
COM 3	Port 1	V90S AIS Data	192.168.10.25	38400

SATELLITE COMMUNICATIONS

There are two satellite communication systems installed. These are a TracPhone V7-HTS and a Thales VesseLINK 350. The V7-HTS is the primary means of satellite communications with the Thales being the secondary back up system.

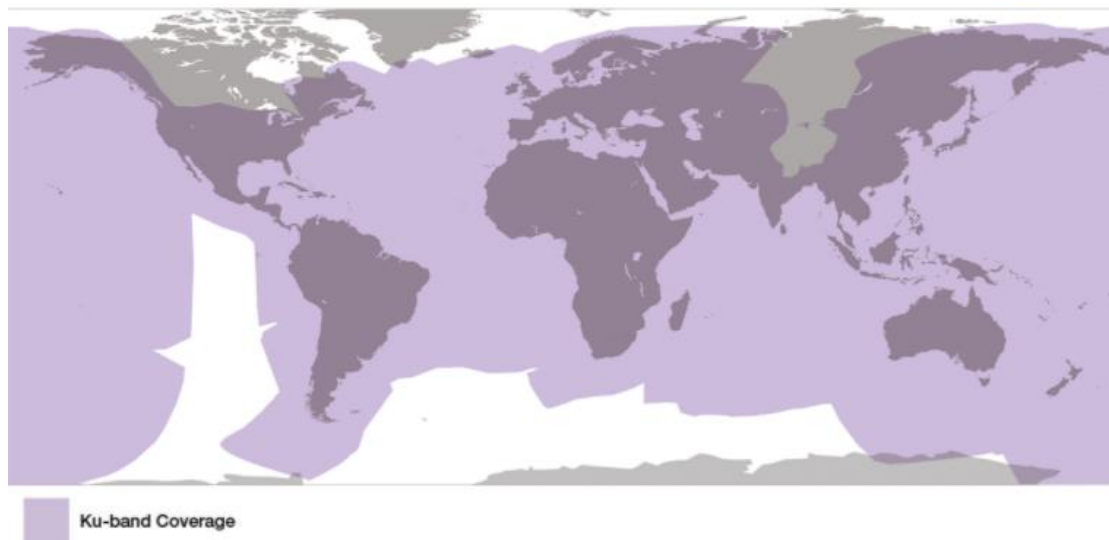
TRACPHONE V7-HTS



TracPhone V7-HTS

KVH's award-winning TracPhone V7hts provides the uncompromised speed and coverage that commercial vessels and yachts worldwide rely on to stay connected at sea. At 60 cm, this rugged, marine-grade antenna outperforms the leading competing 1-meter Ku-band systems and delivers speeds as fast as 10/3 Mbps (down/up) via the layered mini-VSAT Broadband HTS network, dual channels, and powerful tools to manage your data use.

- State-of-the-art VSAT antenna and belowdecks hardware
- High-performance, three-axis stabilized design for superior tracking
- Exclusive dual channel configuration, with hybrid airtime plans that deliver an unlimited use data channel running at the same time as the high-speed channel
- Global layered HTS network with more than 272 million sq. km (105 million sq. miles) coverage
- • Access to the secure myKVH Portal and powerful mini-VSAT Manager data and network management tools
- Integrated network cybersecurity measures



V7-HTS Coverage

The system is composed of an Integrated Commbox Modem (ICM) and the V7-HTS dome. The ICM is mounted on the lazarette electronics panel and the dome is mounted on the antenna frame. There are two LMR400-75 coax cables and a multicore data cable linking the ICM and dome.

It has been requested by the client that the dome is removable. To accommodate this a junction box and an extra pair of coax connectors have been installed at the base of the antenna frame. The junction box is used for the data cables and the F-type connectors for the coax cables. In addition, the V7-HTS cables have a separate cable run through the frame for easy removal.

The ICM is powered by an AC supply at the lazarette panel. The handset is located at the navigation station and is wired into the ICM using RJ11 cable and connectors. Both the high speed and unlimited data channels have been used and hence there are two WAN connections to the Peplink HD2.

The webserver page of the V7 HTS can be accessed by from the network on **192.168.6.1**

This will only work if the WAN1 and WAN2 connections on the HD2 Dashboard have been placed in priority 1. An alternative way to access the webpage of the V7 HTS is to connect to the internal Wi-Fi of the V7.

SSID: KVH-1080900572 2

Password: password

Login detail are Administrator/password

No transmit zones have been enabled to protect crew from the antennas radiated RF energy.

These are as followed:

Azimuth no transmit range: 270 through to 90.

Elevation no transmit range: -25 to 45

In order to **make a call** using the handset 011 has to be dialled first following the number with the international code without the 0's.

For example, to dial a Spanish mobile number you would normally dial 0034 followed by the number. On the V7 HTS you have to dial 01134 followed by the number.

Some system details can be seen below:

Software Versions

Operating System	201
ICM/Antenna	3020
Modem	2.0.1.0-1843
MTA	9.15.34
CommBox	1.22.1

Hardware Information

Antenna S/N	181001087
ICM S/N	180900572
Modem S/N	68346

Modem DID	335612666
MTA MAC Address	00:10:99:12:58:75

THALES VESSELINK 350

Thales VesseLINK provides global satellite coverage for maritime communications. With uninterrupted coverage from pole to pole, it is the solution to depend on for essential communications

VesseLINK operates using Iridium Certus broadband services over a network of 66 satellites that cover 100% of the globe. The solution utilizes this robust network service to provide highly reliable, mobile and essential voice, text and web communications.



Thales VesseLINK 350

The system is composed of a Below Deck Unit (BDU) and an Above Deck Unit (ADU)/Antenna. The BDU is located at the lazarette equipment panel and the antenna is mounted on the antenna frame. They are linked with a single LMR300 coaxial cable. The BDU is powered by a 24V DC supply. A Thales SureLINK IP phone has been installed at the nav station. Note that the IP handset is not directly connected to the LAN port of the Thales BDU but rather through the HD2 at the lazarette and the SD8 switch at the nav station. This connection is on its own dedicate VLAN.

The HD2 WAN interface does allow for the Thales to be chosen as source of internet but only over Wi-Fi WAN.

Both Intel NUC computers at the nav station have a dedicated ethernet connection to the Thales VesseLINK. The ethernet adapters for this are named Iridium. Again, these connections are on a dedicated VLAN.

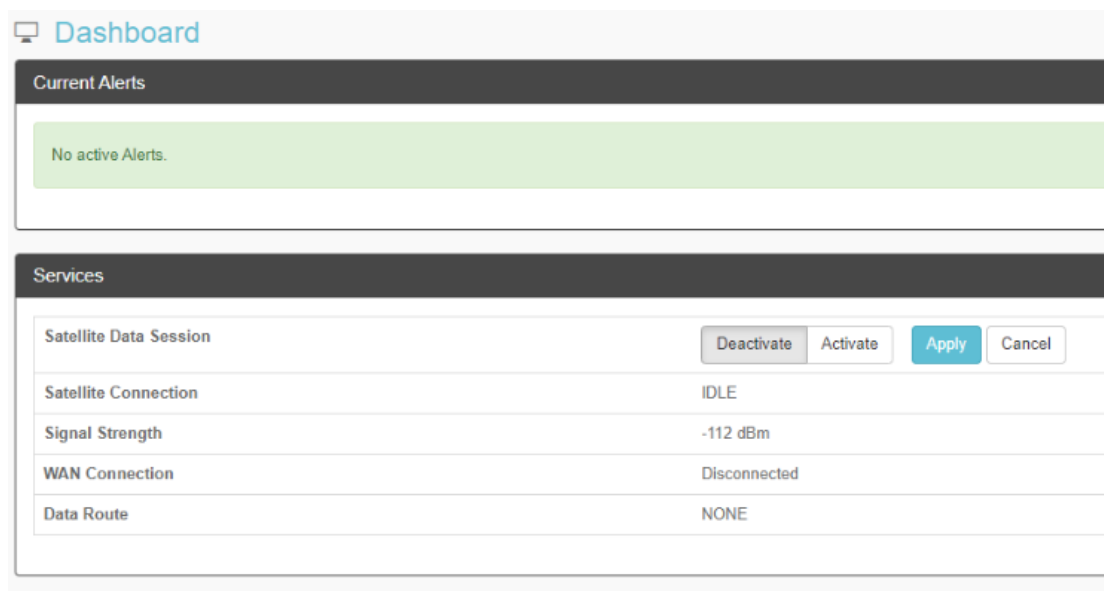
The webpage of the Thales can be accessed from both nav station computers on **192.168.55.1**

The webpage can also be accessed using the internal Wi-Fi of the Thales unit. Details are as follows:

SSID: ThalesLINK
Password: password

Once in the webpage login details are **admin/admin**

To start a Satellite Data Session this must be activated on the main webpage dashboard.



To dial a number using the IP handset you have to dial 9 followed by the international code without the 0's followed by the number itself.

IRIDIUM 9575

An Iridium 9575 handset has also been supplied. This will be stored in the grab bag and used in the event of emergency.

At the time of commissioning the SIM card had not been activated.

INTERCOMS

A custom intercoms solution by Farallon Electronics has been installed to allow for communication between the nav station and the two helm stations.

The system is composed of the main intercoms board which is mounted inside an enclosure at the nav station and two-pre-amp boards which are located at the helm stations.

Please refer to schematics for full details.

At each helm there is a microphone, speaker, and signal button. At the nav station there is a microphone, push to talk button (PTT), LED and four rotary controls.

These rotary controls are:

- Deck Mic Gain Port
- Deck Mic Gain Stbd
- Signal Alert Volume
- Deck Speaker Volume

The deck microphones are always live, and the volume of each mic can be controlled from the nav station. If the nav station isn't paying attention, then the person helming can press a signal button that triggers a buzzer and a led light at the nav. The person at the nav station can then turn up the relevant volume controls and communicate. While the deck microphones are always live the nav station one is push to talk.

A set of spare boards have been provided in the unlikely event of a failure.

IP ADDRESS LIST

Device	IP Address / VLAN
HD2 - Lazarette	192.168.2.1
Switch #1 – Nav Station	192.168.2.51
Switch #2 – Nav Station	192.168.2.50
Switch #3 – Mast Base	192.168.2.52

V7 HTS	192.168.6.1
Iridium Thales	192.168.55.1
PC#1	192.168.10.22
PC #2	192.168.10.21
FARO	192.168.10.10
ATP1 CPU	192.168.10.54
Quadrans	192.168.10.12
MOXA NPort (Nav Station)	192.168.10.23
MOXA NPort (Mast Base)	192.168.10.25
Pixel Sur Mer	192.168.10.18 / 192.168.10.19
ETA Computer / Webserver	10.3.10.11 / 10.3.10.12
Cariboni	192.168.10.11
Panasonic Toughpad	192.168.10.34

WIRING SCHEDULE

The wiring schedule can be found in a separate document accompanying the systems guide.

MANUALS

Product / Manual Title	Link
Pepwave Manuals	https://www.peplink.com/support/downloads/
V90s VHF Manual	https://ww2.bandg.com/downloads/v90-owners-manual/
FARO Manager User Guide	https://irp-cdn.multiscreensite.com/2f2908bd/files/uploaded/Faro_Manager_Users_Guide_v2.0.pdf
KVH V7 HTS	TracPhone V7-HTS User's Guide (groundcontrol.com)
Thales VesseLink	https://www.groundcontrol.com/Iridium_Certus/Thales_Vesselink_User_Manual.pdf
A+T User Guides	Downloads - A and T Instruments